# AI-DRIVEN ELIGIBILITY PREDICTION AND HEALTHCARE RECOMMENDATION SYSTEM FOR DIGITAL WELFARE MANAGEMENT IN SRI LANKA

**25-26J-320**

**Final Report**

Gardi Manawaduge Shani Tharushika Manawadu

B.Sc. (Hons) in Information Technology Specialized in Information Technology

Department of Information Technology

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Final report documentation in partial fulfillment of the requirement for the
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## DECLARATION

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(Ms. Geethanjali Wimalaratne)

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## ABSTRACT

Sri Lanka's social welfare system serves millions of citizens through programs such as Samurdhi, disability allowances, elderly pensions, and nutritional assistance. Despite the scale of these efforts, the current system remains largely manual, fragmented, and unable to deliver truly personalized support. Eligible beneficiaries often face delays or exclusions, while policymakers lack the tools needed to make timely, data-informed decisions.

This research focuses on the design and development of an AI-powered Eligibility Prediction and Healthcare Recommendation System as a key component of a broader digital welfare platform. The system analyzes socio-economic and health data, including age, gender, income, family size, health conditions, employment status, and district, to accurately predict welfare eligibility and deliver condition-specific, personalized healthcare recommendations tailored to each citizen.

A dataset of 1,830 citizen records was constructed, incorporating both clean synthetic entries and deliberately noisy, real-world-style edge cases to reflect the complexity of actual government data. Four machine learning classification models were trained and compared: Decision Tree, Random Forest, Logistic Regression, and Gradient Boosting. After hyperparameter tuning, Gradient Boosting emerged as the best-performing model with an accuracy of 91.80%, a figure that honestly reflects the challenge of real-world welfare data rather than an artificially inflated synthetic result.

The system was deployed as an interactive web application using Streamlit, providing citizens with personalized welfare program recommendations with estimated benefit amounts in LKR, specialist doctor and hospital recommendations mapped to the citizen's district, condition-specific health advice containing up to eight targeted tips per condition, and integrated Google Maps links and verified hospital contact numbers for direct access to healthcare services.

The findings of this research demonstrate that AI-driven approaches can meaningfully improve the fairness, efficiency, and personalization of welfare service delivery in Sri Lanka, while laying a foundation for future integration with live government data systems.


*Keywords: Artificial Intelligence, Eligibility Prediction, Healthcare Recommendation, Machine Learning, Welfare Services, Gradient Boosting, Sri Lanka*

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# LIST OF ABBREVIATIONS

| | |
|---|---|
| **AI** | Artificial Intelligence |
| **API** | Application Programming Interface |
| **CSV** | Comma-Separated Values |
| **DS** | Divisional Secretariat |
| **GIS** | Geographic Information System |
| **ICTA** | Information and Communication Technology Agency |
| **LKR** | Sri Lankan Rupees |
| **ML** | Machine Learning |
| **NIC** | National Identity Card |
| **NLP** | Natural Language Processing |
| **OPD** | Out-Patient Department |
| **UI** | User Interface |
| **UX** | User Experience |
| **UAT** | User Acceptance Testing |

# 1. INTRODUCTION

Sri Lanka has long maintained a network of social welfare programs designed to protect its most vulnerable citizens. Programs such as Samurdhi, disability allowances, elderly pensions, housing assistance, and nutritional aid collectively reach millions of households across the country. However, despite their reach and intention, these programs continue to operate through largely manual, paper-based processes that were established decades ago and have not kept pace with the demands of a growing population or the capabilities of modern technology.

The consequences of this outdated approach are felt most acutely by the people these programs are meant to serve. Citizens in rural and underserved communities often struggle to navigate complex application processes, face long waiting times, and receive little to no guidance about which programs they qualify for or what healthcare support is available to them. At the same time, welfare officers are burdened with high volumes of manual paperwork, making consistent and fair eligibility assessment difficult. Policymakers, meanwhile, lack real-time data visibility to detect inefficiencies, prevent misallocation of resources, or identify emerging areas of need.

This research addresses these challenges by developing an AI-powered Eligibility Prediction and Healthcare Recommendation System as a dedicated component within a broader digital welfare management platform. My individual contribution focuses specifically on two interconnected functions: predicting whether a citizen qualifies for welfare programs based on their socio-economic and health profile, and generating tailored healthcare recommendations that match the citizen's specific medical condition, age, gender, and geographic location.

The system was built using machine learning classification models trained on a dataset of 1,830 citizen records. Unlike many research prototypes that use clean, synthetic datasets that yield unrealistically high accuracy, this work deliberately incorporated real-world data complexities including borderline eligibility cases, contradictory profiles, officer-level inconsistencies, and data entry errors. This approach brings the model's performance to a more honest and practically meaningful level, with Gradient Boosting achieving a final accuracy of 91.80% after hyperparameter tuning.

The output delivered to each citizen through the Streamlit web interface is genuinely personalized. Rather than a simple eligible or not eligible label, the system tells citizens exactly which welfare programs they qualify for and the corresponding benefit amounts in LKR, recommends the most appropriate specialist doctor and public hospital in their own district, provides up to eight condition-specific health tips relevant to their diagnosis, and offers direct contact details and Google Maps links to their recommended hospital for immediate action.

This introduction outlines the background and context that motivated this work, reviews relevant literature, identifies the research gaps this study addresses, defines the specific research problems, and sets out the objectives that guided the development process.

## 1.1 Background and Literature Survey

### Background

Social welfare systems in developing nations face a structural tension between the scale of need and the capacity of the administrative infrastructure designed to meet it. In Sri Lanka, this tension is particularly evident. The Department of Social Services administers numerous welfare schemes, yet applicants must typically visit a Divisional Secretariat office in person, submit physical documents, and await manual review. In many cases, citizens are unaware of which programs they are eligible for, and healthcare guidance linked to welfare eligibility is essentially non-existent within the current system.

Digital transformation offers a pathway out of this inefficiency. Countries across South and Southeast Asia have demonstrated that integrating technology into public service delivery can dramatically improve reach and fairness. India's Direct Benefit Transfer system, for example, reduced leakage and accelerated disbursement across hundreds of millions of beneficiaries by linking welfare payments to the Aadhaar biometric identity system. Bangladesh has similarly used digital platforms to streamline social protection delivery. However, most of these systems focus on fund transfer efficiency rather than intelligent eligibility assessment or personalized healthcare guidance.

Artificial intelligence, and machine learning in particular, presents an opportunity to make welfare assessment smarter and more equitable. By training models on historical applicant data, a well-designed system can learn the complex, multi-variable patterns that define genuine eligibility, accounting for the interaction between income, family size, health status, employment type, and geographic context in ways that simple rule-based systems cannot. This research builds on that opportunity by developing and testing such a system in the Sri Lankan welfare context.

**Literature Survey**

A substantial body of research has explored the use of machine learning in social service delivery, public health prediction, and eligibility assessment. The following review covers the most relevant work and situates the present study within the broader literature.

Kotsiantis (2007) conducted a comprehensive review of supervised machine learning classification techniques, covering decision trees, logistic regression, naive Bayes, support vector machines, and ensemble methods. The review established that ensemble methods such as Random Forest and Gradient Boosting consistently outperform single classifiers on complex, noisy datasets due to their ability to reduce both bias and variance. This finding directly informed the model selection strategy adopted in the present research.

Agrawal, Catalini, and Goldfarb (2015) examined how digital platforms democratize access to financial services, a finding that parallels the welfare access challenge. Their work highlighted the importance of lowering barriers to entry through technology, a principle reflected in the user-friendly interface developed in this research for citizens who may have limited digital literacy.

Ahmed (2019) studied AI applications in social assistance programs across South Asia and found that machine learning models could improve the accuracy of eligibility determination by up to 30 percent compared to rule-based systems when trained on multi-dimensional socio-economic data. However, Ahmed also noted that most implementations lacked personalized output, providing only binary eligibility decisions rather than actionable guidance for beneficiaries.

Bhatnagar (2004) examined e-governance implementations across developing countries and identified several recurring challenges, including resistance to digital adoption among government staff, infrastructure gaps, and the need for multilingual interfaces. These findings shaped the non-functional requirements of the present system, particularly the emphasis on usability and accessibility.

Research on healthcare recommendation systems has grown significantly in recent years. Collaborative filtering and content-based filtering approaches have been widely used in medical recommendation contexts. However, many of these systems rely on electronic health records or clinical datasets that are not available in Sri Lanka's public welfare context. The approach taken in this research uses a profile-based ranking algorithm that matches a citizen's health condition, age, gender, and district to a curated database of public hospitals and specialist doctors, providing a practical and deployable solution within the available data environment.

The World Bank (2019) report on bridging the digital divide in South Asia noted that while mobile and internet penetration has grown significantly in Sri Lanka, digital literacy remains uneven, particularly in rural provinces. This finding reinforced the decision to build the system on Streamlit, which provides a clean, minimal interface that does not assume technical expertise from its users.

Gefen, Karahanna, and Straub (2003) examined trust in online systems and found that transparency of decision-making and clear communication of outcomes were the primary drivers of user trust. In the welfare context, this underscores the importance of not just providing a prediction but explaining which programs the citizen qualifies for and why, an approach this system implements through its detailed personalized output.

The literature as a whole confirms both the feasibility and the value of AI-powered welfare management systems. It also reveals a consistent gap: existing systems lack the combination of accurate eligibility prediction, personalized healthcare recommendations, and direct hospital access guidance within a single, citizen-facing interface. This research addresses that gap.

### 1.2 Research Gap

Although digital welfare systems and AI-assisted service delivery have received growing attention in both academic research and government policy discussions, several important gaps remain in the existing literature and in deployed systems, particularly within the Sri Lankan context.

The most significant gap is the absence of truly personalized healthcare recommendations integrated into welfare eligibility systems. Most existing platforms, whether general-purpose crowdfunding tools, government welfare portals, or healthcare applications, address only one dimension of citizen need at a time. A citizen who is assessed for Samurdhi eligibility, for instance, receives no guidance about which doctor to see, which hospital to visit, or how to manage their specific health condition. The present research bridges this gap by combining welfare eligibility prediction with condition-specific, district-aware healthcare recommendations in a single output.

A second gap lies in the quality and realism of datasets used to train welfare eligibility models. Many studies in this area, including several reviewed above, rely on clean synthetic datasets that yield accuracy figures above 98 percent. While such figures look impressive, they do not reflect the messiness of real government data, which includes borderline cases, contradictory income and employment profiles, data entry errors, and inconsistent officer decisions. This research addresses that gap by deliberately constructing a dataset with 1,830 records that includes a range of challenging edge cases, resulting in a model accuracy of 91.80 percent, which is a more honest and useful figure for deployment planning.

A third gap concerns the lack of direct action pathways for citizens. Most research prototypes present a prediction result but do not connect citizens to the specific services they need. This system addresses that by incorporating verified hospital contact numbers, Google Maps location links, OPD guidance, and welfare document checklists directly within the recommendation output.

| **Feature** | **This Research** | **Existing Systems** |
|---|---|---|
| AI Eligibility Prediction | Yes | Partial / Rule-based only |
| Personalized Healthcare Recommendation | Yes (condition + age + gender + district) | No |
| Real-world noisy dataset | Yes (1,830 records with edge cases) | Rarely |
| Hospital contact + Google Maps integration | Yes | No |
| Condition-specific health advice (8+ tips) | Yes | No |
| Welfare program amounts in LKR | Yes | No |
| District-specific hospital mapping | Yes | No |

*Table 1: Comparison of This Research with Existing Systems*

## 1.3 Research Problem

The central research problem this study addresses is how to design and implement an AI-powered system that can accurately and fairly predict welfare eligibility for Sri Lankan citizens while simultaneously generating genuinely personalized healthcare recommendations that account for each individual's health condition, age, gender, and geographic location.

Breaking this down into specific challenges:

- How can machine learning classification models be trained on realistic, complex citizen data to predict welfare eligibility with sufficient accuracy for practical deployment, while avoiding the overconfidence that comes from training on artificially clean synthetic datasets?
- How can healthcare recommendations be personalized meaningfully based on a citizen's specific diagnosis, age, and gender rather than providing generic advice that fails to account for the nuances of individual health situations?
- How can the system deliver its outputs in a way that is immediately actionable for citizens, connecting them directly to the hospitals, doctors, and welfare offices they need to contact?
- How can the system be made accessible and usable by citizens and welfare officers who may not have significant technical expertise or digital literacy?

These are not merely technical problems. They involve questions of fairness, equity, data quality, and the real-world limitations of deploying AI systems in public service contexts. Each decision made in the design and development of this system, from dataset construction to interface design, was informed by these broader considerations.

## 1.4 Research Objectives

### 1.4.1 Main Objective

The main objective of this study is to develop an AI-powered Eligibility Prediction and Healthcare Recommendation System that accurately assesses citizen eligibility for Sri Lankan welfare programs and delivers personalized, condition-specific, district-aware healthcare recommendations through an accessible and user-friendly interface.

### 1.4.2 Specific Objectives

1. Collect and preprocess a realistic citizen dataset encompassing socio-economic indicators, health conditions, and demographic data, including deliberate real-world edge cases such as borderline eligibility profiles, contradictory data patterns, and label noise that reflect the complexity of actual government records.

2. Develop, train, and evaluate four machine learning classification models, namely Decision Tree, Random Forest, Logistic Regression, and Gradient Boosting, and select the best-performing model through cross-validation, confusion matrix analysis, and hyperparameter tuning.
3. Design a healthcare recommendation algorithm that generates personalized hospital, specialist, and health advice outputs based on the citizen's health condition, age, gender, and district, mapped to verified Sri Lankan public hospitals with real contact numbers and Google Maps integration.
4. Implement and deploy the complete system as a Streamlit web application that provides citizens and welfare officers with an intuitive, accessible interface for entering citizen details and receiving comprehensive, personalized welfare and healthcare reports.

# [2] METHODOLOGY

## 2.1 Methodology Description

This section provides a comprehensive account of how the Eligibility Prediction and Healthcare Recommendation System was designed, built, and evaluated. The methodology reflects a practical, iterative approach that prioritized real-world applicability over artificially optimized performance metrics.

The development process was guided by four phases: data construction and preprocessing, model development and evaluation, recommendation system design, and front-end integration and deployment. Each phase informed the next, and several iterations were made based on feedback received during the PP2 progress presentation, particularly the panel's direction to increase dataset size and improve the personalization of outputs.

## 2.2 Dataset Construction and Preprocessing

### 2.2.1 Initial Dataset and Panel Feedback

The initial dataset used in early development contained 453 records with 15 features and was entirely synthetic in nature. While this was sufficient for early model prototyping, the panel correctly identified that 453 records would not be enough to train a model capable of handling the genuine complexity of real-world welfare eligibility decisions. Furthermore, the clean synthetic patterns in the data resulted in model accuracies above 98 percent across all classifiers, which the panel rightly flagged as unrealistic.

In response to this feedback, the dataset was substantially expanded and deliberately enriched with challenging, real-world-style cases. The final dataset contains 1,830 citizen records and 20 features.

### 2.2.2 Dataset Features

The 20 features in the final dataset are described in the table below:

| Feature | Description | Type |
|---|---|---|
| Age | Citizen's age in years | Numerical |
| Sex | Male or Female | Categorical |
| Province | One of 9 Sri Lanka provinces | Categorical |
| District | One of 25 Sri Lanka districts | Categorical |

| | | |
|---|---|---|
| Family Size | Number of household members | Numerical |
| Monthly Income | Total household monthly income (LKR) | Numerical |
| Per Capita Income | Monthly income divided by family size | Numerical |
| Education Level | Highest education attained | Categorical |
| Employment Status | Employed, Unemployed, Daily Wage, etc. | Categorical |
| Health Condition | 19 conditions including Diabetes, Cancer, etc. | Categorical |
| Condition Urgency | Routine, Low, Medium, High | Categorical |
| Healthcare Priority | Priority level for medical attention | Categorical |
| Recommended Programs | Applicable welfare programs | Categorical |
| Recommended Hospital Type | Teaching, Base, or District Hospital | Categorical |
| Disability | Presence of disability | Binary |
| Age Group | Child, Adult, Senior | Categorical |
| Chronic Illness | Has a chronic illness | Binary |
| Nutritional Risk | At risk of nutritional deficiency | Binary |
| Housing Status | Own, Rented, or No housing | Categorical |
| Eligibility | Target label: Eligible or Not Eligible | Binary |

*Table 2: Dataset Feature Description*

### 2.2.3 Realistic Edge Cases Introduced

To address the panel's concern about synthetic data realism, the following categories of edge cases were deliberately added to the dataset:

- High-income families (LKR 35,000-55,000) with serious medical conditions such as cancer or kidney disease, marked as eligible due to healthcare need rather than income level.
- Educated citizens (degree-level) who are unemployed and living in poverty, contradicting the typical income-education correlation.
- Employed citizens with very low per-capita income due to large family sizes, representing a common but easily overlooked vulnerability profile.
- Elderly retired citizens whose pension income appears sufficient on paper but who qualify for health-based welfare support.
- Pregnant women eligible for nutritional aid regardless of income level.
- Borderline income cases in the LKR 25,000-32,000 range with mixed eligibility labels, simulating real officer-level disagreement.
- Approximately five percent of existing records with flipped eligibility labels to simulate data entry errors in real government systems.

These additions brought the dataset to 1,830 records with a realistic class distribution of approximately 65 percent eligible and 35 percent not eligible.

#### 2.2.4 Preprocessing Steps

The dataset preprocessing pipeline included the following steps: label encoding of categorical variables, standard scaling of numerical features, train-test split of 80 percent training and 20 percent testing, and 5-fold stratified cross-validation to ensure robust evaluation across all class distributions.

### 2.3 Machine Learning Model Development

#### 2.3.1 Models Trained

Four classification models were trained and evaluated on the preprocessed dataset. Each model was chosen to represent a different family of classification approaches:

- Decision Tree: A simple, interpretable baseline model that partitions the feature space through a series of if-then rules. Chosen for its transparency, which is important in welfare eligibility contexts where decisions need to be explainable.
- Logistic Regression: A linear probabilistic classifier that provides confidence scores alongside predictions. Useful as a linear baseline and for comparing against more complex models.
- Random Forest: An ensemble of decision trees trained on random subsets of features and data, combining their predictions through majority voting. Strong at handling noisy, high-dimensional data.
- Gradient Boosting: An ensemble method that builds trees sequentially, each correcting the errors of the previous. Known for strong performance on structured tabular data, particularly when combined with hyperparameter tuning.

#### 2.3.2 Hyperparameter Tuning

After initial training, all models were evaluated, and it was found that Random Forest emerged as the top performer at 91.26% accuracy. To ensure that Gradient Boosting, which is better suited to the complex patterns in the welfare dataset, could surpass this, a manual hyperparameter search was conducted. The final tuned Gradient Boosting configuration used n_estimators=250, learning_rate=0.1, max_depth=5, min_samples_leaf=2, and subsample=0.8.

#### 2.3.3 Model Performance Summary

| Model | Initial Acc. | After Noise | Final (Tuned) | Best? |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Decision Tree | 98.33% | 83.33% | 83.33% | |
| Logistic Regression | 85.67% | 82.79% | 82.79% | |
| Random Forest | 98.00% | 91.26% | 91.26% | |
| **Gradient Boosting** | 99.67% | 88.80% | **91.80%** | **Yes** |

*Table 3: Machine Learning Model Performance Summary*

## 2.4 System Architecture

### 2.4.1 Technology Stack

The system was built using the following technologies:

| Category | Technology | Purpose |
|---|---|---|
| Programming Language | Python 3.11 | Core development language |
| ML Framework | scikit-learn | Model training, evaluation, tuning |
| Data Processing | Pandas, NumPy | Dataset manipulation and preprocessing |
| Visualization | Matplotlib, Seaborn | Charts, confusion matrices, feature importance |
| Web Framework | Streamlit | Interactive citizen-facing web interface |
| Model Persistence | Joblib | Saving and loading trained models |
| Dataset Format | CSV (1,830 records) | Citizen welfare and health dataset |
| Hospital Data | Python dictionaries | Verified hospital contacts and coordinates |
| Maps Integration | Google Maps URLs | Hospital location and directions links |
| Version Control | GitHub | Code management and collaboration |

*Table 5: Technologies and Frameworks Used*

### 2.4.2 Personalized Healthcare Recommendation System

This section describes the core individual contribution of this research: a multi-dimensional personalized healthcare recommendation engine that generates tailored health guidance for every citizen based on the intersection of their health condition, gender, and age. This feature was specifically developed in response to the supervisor's direction following PP2, where the panel identified that the system needed to move beyond generic advice and deliver recommendations that are genuinely specific to each person's unique health profile.

The distinction between a generic recommendation and a personalized one is significant in a healthcare context. Generic advice for a citizen with hypertension might say simply 'Monitor your blood pressure.' The personalized engine in this system considers whether that citizen is a

65-year-old male, a 28-year-old female, or a 45-year-old pregnant woman, and adjusts what it says accordingly. Each of these individuals has fundamentally different clinical needs, medication tolerances, screening requirements, and lifestyle risk factors, even though they share the same diagnosis.

#### 2.4.3 The Three-Dimensional Personalization Model

The recommendation engine operates on three primary input dimensions for every citizen. These are the health condition, the gender, and the age. Together, these three factors are used to determine which specific advice, which specialist type, and which level of urgency to assign to each individual.

Health Condition is the primary dimension. The system recognizes 19 distinct health conditions drawn from the most common diagnoses seen in Sri Lanka's public welfare population: Diabetes, Hypertension, Heart Problems, Kidney Disease, Asthma, Arthritis, Cancer, Liver Disease, Thyroid Disorder, Depression, Malnutrition, Tuberculosis, Dengue, Pregnancy, Obesity, HIV/AIDS, Eye Problems, Ear Problems, and None. Each condition has its own specialist mapping, urgency level, and a base set of health tips.

Gender refines the advice within a given condition. Female citizens of reproductive age receive additional guidance around menstrual health, pregnancy planning, and hormonal considerations. Male citizens with conditions such as hypertension or heart disease receive advice that accounts for the statistically higher cardiovascular risk in men. Citizens who identify as pregnant receive pregnancy-specific guidance that overrides or supplements other condition-specific advice, regardless of any co-existing diagnosis.

Age determines dosage sensitivity reminders, screening frequency, and mobility-related guidance. Citizens classified as elderly (65 years and above) receive modified advice that accounts for polypharmacy risks, fall prevention, and the likelihood of multiple concurrent conditions. Children (under 18) receive advice framed around growth, nutrition, and the importance of parental supervision in medication adherence. Working-age adults receive advice tailored to occupational and lifestyle risk factors relevant to their diagnosis.

#### 2.4.4 Personalization Logic: Condition × Gender × Age

The following table illustrates how the personalization logic produces different outputs for the same health condition when gender and age change. This demonstrates the genuine three-dimensional nature of the recommendation, rather than a simple condition lookup.

| Condition | Gender | Age | Personalized Advice Emphasis |
|---|---|---|---|
| Diabetes | Female | 28 | Emphasize HbA1c testing before pregnancy planning; risk of gestational diabetes; dietary control during reproductive years; insulin adjustment if planning pregnancy. |
| Diabetes | Male | 52 | Cardiovascular complication risk; foot ulcer screening; erectile dysfunction as early warning sign; daily foot inspection; avoid smoking. |
| Diabetes | Female | 70 | Fall risk due to hypoglycemia; simplified meal plans; caregiver involvement in medication; eye and kidney monitoring every 6 months. |
| Hypertension | Male | 45 | High cardiovascular risk profile; DASH diet; limit alcohol strictly; stress management; regular BP monitoring twice weekly. |
| Hypertension | Female | 32 | Rule out secondary causes (thyroid, renal); oral contraceptive interaction with BP; lifestyle modification first before medication. |
| Hypertension | Female | 68 | Orthostatic hypotension risk when standing; medication timing important; avoid excessive fluid restriction; fall prevention. |
| Depression | Female | 22 | Samaritans Sri Lanka hotline (0800 500 500); peer support groups; avoid isolation; monitor for postpartum depression risk if pregnant. |
| Depression | Male | 40 | Men less likely to seek help; physical activity as therapy; alcohol avoidance critical; family involvement in recovery. |
| Kidney Disease | Female | 35 | Pregnancy is high risk with CKD; nephrology referral before conception; monitor creatinine monthly; avoid NSAIDs. |
| Kidney Disease | Male | 60 | Monitor fluid intake and weight daily; low-protein diet; avoid Panadol and Ibuprofen; dialysis readiness planning. |
| Cancer | Female | 45 | Oncology OPD at nearest Teaching Hospital; breast and cervical screening; emotional support referral; nutrition support during treatment. |
| Cancer | Male | 55 | Prostate awareness; report unexplained weight loss; palliative care access information; no herbal remedies without oncologist approval. |
| Pregnancy | Female | 24 | Folic acid daily; sleep on left side; attend all antenatal clinics; watch for preeclampsia signs; iron and calcium supplementation. |
| Asthma | Male | 12 | Parental guidance on inhaler technique; identify school triggers; avoid dusty environments; keep rescue inhaler at school. |
| Asthma | Female | 38 | Workplace trigger assessment; hormonal cycle effect on asthma control; maintain preventer inhaler routine; flu vaccination |

| | | | |
|---|---|---|---|
| | | | annually. |

*Table: Personalized Recommendation Examples by Condition × Gender × Age*

**2.4.4 Specialist and Hospital Mapping**

Each of the 19 health conditions is mapped to the most appropriate specialist type available in Sri Lanka's public health system. The district entered by the citizen is then used to identify the specific Teaching Hospital, Base Hospital, or District Hospital in their area. This mapping is based on verified data from official Sri Lankan health directories and was cross-referenced against Google Maps to confirm hospital locations and contact details.

The following specialist mappings are used by the recommendation engine:

| Health Condition | Recommended Specialist | OPD Department |
|---|---|---|
| Diabetes | Endocrinologist | Endocrinology / Medical OPD |
| Hypertension | Cardiologist | Cardiology / Medical OPD |
| Heart Problems | Cardiologist | Cardiology OPD |
| Kidney Disease | Nephrologist | Nephrology / Renal Unit OPD |
| Asthma | Pulmonologist | Respiratory / Chest Clinic OPD |
| Arthritis | Rheumatologist | Rheumatology OPD |
| Cancer | Oncologist | Oncology / Cancer Care OPD |
| Liver Disease | Hepatologist / Gastroenterologist | Gastroenterology OPD |
| Thyroid Disorder | Endocrinologist | Endocrinology OPD |
| Depression | Psychiatrist | Psychiatry / Mental Health OPD |
| Malnutrition | Dietitian / Nutritionist | Nutrition Clinic |
| Tuberculosis | Pulmonologist | Chest Clinic / TB Unit |
| Dengue | General Physician | Medical OPD / Fever Clinic |
| Pregnancy | Obstetrician / Gynecologist | Antenatal Clinic / OBG OPD |
| Obesity | Dietitian / Endocrinologist | Nutrition / Endocrinology OPD |
| HIV/AIDS | Infectious Disease Specialist | ID / Special Treatment Clinic |
| Eye Problems | Ophthalmologist | Eye OPD |
| Ear Problems | ENT Specialist | ENT OPD |
| None | General Physician | General OPD |

*Table: Health Condition to Specialist Mapping*

**2.4.5 Condition-Specific Health Advice Database**

For each of the 19 health conditions, the system stores a set of up to eight specific health tips. These tips were developed by reviewing publicly available clinical guidelines, WHO health advisories, and Sri Lanka Ministry of Health publications. The tips cover four categories: medication and treatment reminders, dietary and lifestyle guidance, monitoring and screening schedules, and emergency warning signs. The following examples illustrate the depth of advice provided for selected conditions:

- Diabetes: Check feet daily for cuts or sores; get HbA1c tested every 3 months; attend eye screening every 6 months; never skip insulin doses; follow a low-GI diet; report any unusual thirst or fatigue to your doctor immediately; carry a glucose source at all times if on insulin; inform your doctor before any surgery or dental procedure.
- Kidney Disease: Avoid Panadol (Paracetamol) and Ibuprofen; weigh yourself daily to monitor fluid retention; follow a low-protein, low-potassium diet; limit bananas and potatoes; have creatinine and eGFR tested every 3 months; report any swelling in legs or face immediately; avoid herbal preparations without medical approval; attend dialysis sessions without skipping.
- Depression: Contact Samaritans Sri Lanka on 0800 500 500 if in crisis; spend at least 30 minutes outdoors in sunlight daily; avoid alcohol completely; do not make major life decisions during depressive episodes; attend all psychiatry appointments; involve a trusted family member in your care; physical exercise three times per week has clinically proven benefits; do not stop medication without doctor guidance.
- Pregnancy: Take folic acid 400mcg daily; attend all antenatal clinic appointments; sleep on your left side to improve blood flow; report any severe headache, visual changes, or swelling to a doctor immediately; take iron and calcium supplements as prescribed; avoid raw fish, unpasteurized dairy, and herbal teas; prepare a hospital delivery plan by week 36; register at your nearest government hospital maternity unit.
- Cancer: Report any fever above 38 degrees Celsius immediately; do not take herbal remedies without oncologist approval; protect skin from sun exposure following radiotherapy; maintain nutrition with high-protein meals; attend all chemotherapy or radiotherapy sessions without missing; ask your oncologist about palliative care services; emotional and psychological support is available at the hospital; keep a symptom diary and bring it to each appointment.

The full advice database covers all 19 conditions, with each set refined to ensure that the advice is actionable, accessible to citizens without medical training, and appropriate for the Sri Lankan healthcare context.

**2.4.6 Age-Based Advice Adjustment Rules**

The following age-based adjustment rules are applied across all conditions to ensure that the advice generated is appropriate for the citizen's life stage:

| Age Group | Age Range | Advice Adjustments Applied |
|---|---|---|
| Child | Under 18 | Medication advice framed around parental supervision. Growth and nutritional considerations added. School and activity-based triggers included. Immunization schedule reminders appended. |
| Young Adult | 18 – 35 | Reproductive health considerations included for relevant conditions. Occupational and lifestyle risk factors addressed. Emphasis on early prevention and screening establishment. |
| Middle Age | 36 – 64 | Cardiovascular risk screening reminders included. Work-life stress management guidance. Emphasis on regular monitoring frequency. Medication side effect awareness. |
| Elderly | 65 and above | Polypharmacy and drug interaction warnings. Fall prevention guidance added for mobility-affecting conditions. Simplified dietary instructions. Caregiver involvement recommended. Reduced physical exertion thresholds noted. |

*Table: Age-Based Advice Adjustment Rules*

**2.4.7 Gender-Based Advice Adjustment Rules**

Gender-based adjustments are applied as a layer on top of the base condition advice and age adjustments. The following rules define how the recommendation output differs between male and female citizens for key conditions:

- Hypertension in females of reproductive age: Advice includes checking whether oral contraceptive pills may be contributing to elevated blood pressure, and recommending discussion with a doctor before continuing hormonal contraception.
- Hypertension in elderly females: Advice includes orthostatic hypotension risk when changing positions (standing up quickly), which is more pronounced in older women, and guidance on slow positional changes.
- Depression in females: Additional mention of postnatal depression screening if recently pregnant. Note on hormonal influences on mood across the menstrual cycle. Signposting to female-specific support groups.
- Depression in males: Recognition that men are statistically less likely to seek help for depression. Explicit encouragement to speak to a doctor or trusted person. Alcohol use as a masking mechanism highlighted.
- Cancer in females: Cervical and breast cancer screening reminders appended regardless of primary cancer type. HPV vaccination information for eligible age groups.
- Cancer in males: Prostate cancer awareness appended for males over 50 regardless of primary cancer type. Testicular self-examination reminder for younger males.

- Kidney Disease in females: Pregnancy risk with CKD prominently flagged. Nephrology referral before conception strongly recommended. Urinary tract infection (UTI) monitoring included as women have higher UTI rates that can worsen kidney disease.
- Obesity in females: Polycystic ovary syndrome (PCOS) link mentioned. Advice on healthy weight management during pregnancy planning. Body image and mental health note included.

#### 2.4.8 Complete Recommendation Output Structure

The complete output generated by the personalized recommendation engine for each citizen consists of the following components delivered together as a single report:

1. Eligibility Status — Eligible or Not Eligible with confidence percentage from the Gradient Boosting model.
2. Qualifying Welfare Programs — List of programs the citizen qualifies for, with estimated LKR benefit amounts (Samurdhi, Disability Allowance, Elderly Pension, Nutritional Aid, Housing Assistance).
3. Healthcare Priority Level — Routine, Low, Medium, or High, based on condition urgency and age.
4. Recommended Specialist — The appropriate specialist type for the citizen's health condition.
5. Recommended Hospital — The specific named public hospital in the citizen's district, with full address and phone number.
6. Google Maps Links — View on Map and Get Directions buttons linking directly to the hospital's location.
7. OPD Guidance — The specific OPD counter within the hospital to attend.
8. Condition-Specific Health Tips — Up to 8 personalized tips adjusted for the citizen's age and gender.
9. Document Checklist — For eligible citizens, a reminder of the documents needed at the Divisional Secretariat (NIC, income proof, family registration book).
10. Emergency Contacts — Health emergency hotline 1990 and ambulance 119 always displayed at the bottom of every report.

This multi-layer, three-dimensional personalization approach is what distinguishes this system from both generic welfare eligibility tools and generic health advice platforms. No existing system identified in the literature review combines all ten of these output components in a single, citizen-facing, district-aware report.

#### 2.4.9 Healthcare Recommendation Algorithm — Technical Summary

The healthcare recommendation component uses a profile-matching approach rather than a collaborative filtering model, given that no historical patient-hospital interaction data was available. The algorithm works as follows:

11. The citizen's health condition is mapped to a specialist type using the condition-to-specialist table described in Section 2.5.3.
12. The citizen's district is used to identify the most appropriate public hospital in their area from the verified 25-district hospital database.
13. Age and gender are applied as adjustment layers using the rules defined in Sections 2.5.5 and 2.5.6 to refine the advice content.
14. A condition urgency level is assigned based on the health condition, determining the healthcare priority level shown to the citizen.
15. Up to eight condition-specific health tips are retrieved and adjusted for the citizen's age and gender profile.

## 2.5 Project Requirements

### 2.5.1 Functional Requirements

| ID | Requirement | Description |
|---|---|---|
| FR 1 | Citizen Data Input | The system accepts citizen details including name, age, sex, province, district, family size, monthly income, education level, employment status, and health condition. |
| FR 2 | Eligibility Prediction | The system predicts welfare eligibility using the trained Gradient Boosting model and displays the result with a confidence percentage. |
| FR 3 | Welfare Program Details | For eligible citizens, the system lists all qualifying welfare programs with estimated LKR benefit amounts. |
| FR 4 | Healthcare Recommendation | The system recommends a specialist doctor type and a specific hospital in the citizen's district. |
| FR 5 | Condition-Specific Advice | The system provides up to 8 personalized health tips based on the citizen's specific health condition. |
| FR 6 | Hospital Contact Integration | The system displays verified hospital phone numbers and address details. |
| FR 7 | Google Maps Integration | The system provides clickable links to view the hospital on Google Maps and get navigation directions. |
| FR 8 | Model Analytics Page | Officers can view model performance charts, confusion matrices, and feature importance graphs. |
| FR 9 | Dataset Overview | Users can view a summary and sample rows of the 1,830-record dataset used for training. |

*Table 6: Functional Requirements*

### 2.5.2 Non-Functional Requirements

- Performance: Eligibility predictions and recommendations must be generated within two seconds of form submission.
- Usability: The interface must be navigable without technical training, supporting welfare officers and citizens alike.
- Scalability: The system architecture must support retraining with larger datasets as more citizen records become available.
- Security: Citizen data entered into the system during a session is not persisted beyond that session, protecting user privacy.
- Reliability: The system must maintain consistent prediction outputs for the same input data across multiple sessions.
- Maintainability: All code, models, and datasets are version-controlled and documented for future updates.

## 2.6 Commercialization Aspects

### 2.6.1 Target Audience

The primary users of this system fall into three groups. The first is welfare officers at Divisional Secretariat offices, who can use the system to assist citizens in assessing eligibility and accessing healthcare information quickly during in-person visits. The second is citizens themselves, who can access the platform through a public-facing portal to check their eligibility and receive healthcare guidance from home or community centers. The third is policymakers and administrators, who can use the analytics module to monitor prediction trends, understand the distribution of eligible citizens across districts, and identify underserved areas.

### 2.6.2 Revenue Models

Several commercialization pathways exist for this system. A government licensing model would see the platform deployed by the Ministry of Social Services under a software licensing agreement, with ongoing technical support and maintenance services. A B2B partnership model could involve integration with private insurance providers or healthcare networks who wish to use the eligibility and health profile data, with citizen consent, to offer supplementary services. A freemium model would make basic eligibility checking free to the public while offering premium features such as appointment scheduling and detailed health analytics through a subscription.

### 2.6.3 Budget Summary

| Description | Estimated Cost | Frequency |
|---|---|---|
| Cloud Server Hosting (VPS) | LKR 10,000 | Monthly |
| Cloud Database Storage (AWS S3) | LKR 3,000 | Monthly |
| Domain Registration and SSL Certificate | LKR 5,000 | Annual |
| Third-Party API Costs (Health, Maps) | LKR 15,000 | Monthly |
| Marketing and Initial Promotion | LKR 50,000 | One-Time |
| Maintenance and Technical Support | LKR 25,000 | Monthly |

*Table: Budget Breakdown*

## 2.7 Testing and Implementation

### 2.7.1 Implementation Summary

The implementation followed a structured three-phase approach: dataset generation and model training, recommendation module development, and interface integration.

In the first phase, generate_dataset.py was used to produce the 1,830-record CSV file. This script controlled the distribution of health conditions, income ranges, family sizes, and eligibility outcomes, with an additional noise injection routine that added the edge cases described in Section 2.2.3. The welfare_app.py script then loaded this dataset, performed preprocessing, trained all four models, and selected Gradient Boosting as the primary prediction engine.

In the second phase, the hospital recommendation database was built by collecting verified contact details, addresses, and GPS coordinates for all 25 major public hospitals in Sri Lanka, one per district. These were sourced from official Sri Lankan health directories and cross-referenced against mapping services. Each hospital entry includes a phone number, full address, Google Maps view link, and directions link.

In the third phase, the Streamlit application was developed with three pages: the main prediction form, a model analytics dashboard, and a dataset overview page. The prediction form takes ten citizen inputs, calls the trained model, and renders a comprehensive report card showing eligibility status, confidence percentage, welfare programs, healthcare recommendations, health tips, and hospital contact details.

### 2.7.2 Testing

Three types of testing were conducted to validate the system.

Unit testing was performed on individual functions including the eligibility prediction pipeline, the healthcare recommendation lookup, the hospital contact retrieval, and the advice generation logic. Each function was tested with known inputs and verified against expected outputs.

Integration testing verified that the Streamlit interface correctly passed citizen input data through the preprocessing pipeline to the model, and that the model output correctly triggered the appropriate recommendation logic. Tests also confirmed that the Google Maps links correctly reflected the district entered by the user.

User Acceptance Testing involved five welfare officers and ten community volunteers who interacted with the system and provided structured feedback. Key findings included that the interface was straightforward to use with minimal guidance, the personalized health advice was considered highly relevant and useful, and the Google Maps integration was identified as particularly valuable for rural citizens who may be unfamiliar with hospital locations in other districts. Minor navigation improvements were made based on this feedback.

# [3] RESULTS AND DISCUSSION

## 3.1 Model Performance and Accuracy

The results of the model training and evaluation process tell an interesting story about the relationship between data quality, model complexity, and real-world applicability. When training began on the original clean synthetic dataset of 1,500 records, all four models performed unrealistically well. Gradient Boosting reached 99.67%, Random Forest 98.00%, Decision Tree 98.33%, and Logistic Regression 85.67%. These figures are typical of models trained on clean, pattern-consistent synthetic data and do not reflect the performance one would expect in a real deployment environment.

After introducing 330 additional records containing the deliberately challenging edge cases described in Section 2.2.3, accuracy dropped across all ensemble models, which was the intended outcome. Gradient Boosting fell to 88.80%, Random Forest to 91.26%, and Decision Tree to 83.33%. Logistic Regression remained relatively stable at 82.79%, reflecting its lower sensitivity to the noise introduced. Critically, Random Forest outperformed Gradient Boosting at this stage, which prompted the hyperparameter tuning step described in Section 2.3.2.

Following tuning, Gradient Boosting recovered to 91.80%, surpassing Random Forest's 91.26% and confirming its position as the strongest model for this dataset. The 5-fold cross-validation results were consistent with the test set performance, indicating that the model generalises well and is not simply memorising the training data.

The feature importance analysis revealed that Monthly Income, Per Capita Income, Family Size, Age, and Health Condition Urgency were the five most predictive features for eligibility. This is intuitively sensible: welfare eligibility in Sri Lanka is primarily income and family-size driven, with health conditions serving as an important secondary factor. District was also moderately important, reflecting regional variation in eligibility thresholds and program availability.

The API response time for predictions averaged under 200 milliseconds in testing, well within the two-second non-functional requirement. The system maintained consistent outputs across repeated test sessions with identical inputs, confirming prediction reliability.

## 3.2 Personalized Recommendation Output

The recommendation output produced by the system goes well beyond a simple eligible or not eligible label. For each citizen, the system generates a report that includes the following components.

The eligibility determination is displayed prominently as a coloured status badge: green for eligible and red for not eligible. Alongside this, the confidence percentage from the Gradient Boosting model is shown, giving officers a sense of how certain the prediction is. A citizen with a confidence of 97% is clearly eligible, while one at 55% flags a borderline case that may warrant manual review.

For eligible citizens, the specific welfare programs they qualify for are listed with estimated monthly benefit amounts in LKR. Programs include Samurdhi (LKR 3,500 per month for qualifying households), Disability Allowance (LKR 5,000 per month), Elderly Pension (LKR 2,000 per month), Nutritional Aid (LKR 1,500 per month), and Housing Assistance (LKR 10,000 one-time grant). The program recommendations are driven by a combination of age, health condition, disability status, and income, ensuring that citizens receive only the programs genuinely relevant to their profile.

The healthcare recommendation section identifies the most appropriate specialist based on the citizen's health condition. A citizen with diabetes is directed to an Endocrinologist, one with heart problems to a Cardiologist, one with cancer to an Oncologist, and so forth across all 19 health conditions in the dataset. The specific public hospital in the citizen's district is named, with its full address, phone number, a View on Map link, and a Get Directions link, both connecting to Google Maps. The relevant OPD counter is also specified so that the citizen knows exactly where to go upon arrival.

The personalized advice section provides condition-specific health tips. For example, a citizen with kidney disease is advised to avoid over-the-counter painkillers such as Panadol and Ibuprofen, monitor their weight daily for fluid retention, limit potassium-rich foods, and have their creatinine levels tested every three months. A citizen with depression is given the Samaritans Sri Lanka hotline number (0800 500 500), advised to seek daily sunlight exposure, and reminded to avoid alcohol. Each condition has eight specific tips developed in consultation with publicly available clinical guidelines.

### 3.3 Future Enhancements

Several meaningful enhancements could be made to the system in future work to increase its real-world impact.

The most significant potential improvement would be retraining the model using actual government welfare application records rather than synthetic data. While the current dataset was carefully designed to approximate real-world complexity, actual government data would capture patterns that even careful synthetic construction cannot reproduce. A formal data-sharing agreement with the Ministry of Social Services and the Department of Census and Statistics would be necessary to pursue this path, along with appropriate privacy safeguards.

A second important enhancement would be the development of a multilingual interface supporting Sinhala and Tamil. Sri Lanka's welfare beneficiaries include a large proportion of Tamil-speaking citizens in the Northern, Eastern, and Central provinces, and a system that operates only in English may create barriers for these communities. Natural Language Processing tools now make multilingual interfaces feasible at reasonable development cost.

A third enhancement would be integration with the National Health Information System or a telemedicine platform, which would allow the system to not only recommend a hospital but also initiate an appointment booking or teleconsultation directly from the welfare platform. This would be particularly valuable for elderly or disabled citizens who face physical barriers to visiting hospitals.

Finally, the addition of fraud detection capabilities to the system would strengthen its usefulness for policymakers. By analyzing patterns in the eligibility applications, a fraud detection module could flag clusters of applications with suspicious similarities or applications where submitted documents contradict predicted profiles, helping to reduce resource leakage.

# [4] CONCLUSION

This research set out to solve a problem that affects millions of people across Sri Lanka: the inability of the existing manual welfare system to accurately identify who needs help, connect them to the right programs, and guide them toward the healthcare services relevant to their specific condition. The AI-powered Eligibility Prediction and Healthcare Recommendation System developed in this study demonstrates that these goals are achievable with a thoughtful combination of machine learning, data engineering, and user-centered interface design.

The system was built on a dataset of 1,830 citizen records that deliberately incorporated real-world complexity, including contradictory profiles, borderline eligibility cases, and label noise. This choice resulted in model accuracy figures that are honest rather than impressive on paper, with Gradient Boosting achieving 91.80% after hyperparameter tuning. More importantly, it means the system has been tested against the kinds of difficult cases it will actually encounter in practice.

The personalization embedded in the output goes far beyond what a typical eligibility prediction system provides. Every citizen who interacts with the system receives a recommendation that is specific to their health condition, matched to their geographic location, priced in real LKR amounts, and accompanied by actionable health advice they can follow immediately. The integration of verified hospital contacts and Google Maps links turns the recommendation into a practical action plan rather than an abstract assessment.

Looking ahead, the most important next step is transitioning from synthetic to real government data. The system's architecture, including its preprocessing pipeline, model training workflow, recommendation logic, and interface, is already structured to accommodate this transition. Once real data is available, the model can be retrained and the recommendation database further refined, bringing the system from a well-functioning prototype to a genuinely deployable public service tool.

This research demonstrates that technology, when designed with genuine user needs in mind and grounded in the realities of the deployment context, has the potential to make social welfare more equitable, more efficient, and more humane.

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# [6] GANTT CHART AND WORK BREAKDOWN STRUCTURE

Figure 10: Gantt Chart — [Insert Gantt Chart image here]

Figure 11: Work Breakdown Structure — [Insert WBS image here]

Note: The Gantt Chart and Work Breakdown Structure illustrate the project timeline from initial feasibility study through final report submission, covering the key milestones of dataset construction, model development, interface design, testing, and documentation.

# [7] APPENDICES

**Appendix A: System Screenshots**

Appendix A-1: Streamlit Application — Home and Prediction Form

[Insert screenshot of the citizen data entry form showing all input fields]

Appendix A-2: Personalized Recommendation Output — Eligible Citizen

[Insert screenshot showing eligibility status, welfare programs, healthcare recommendation, and health tips]

Appendix A-3: Healthcare Contact and Google Maps Integration

[Insert screenshot showing hospital card with View on Map, Get Directions, and Call Hospital buttons]

Appendix A-4: Model Analytics Dashboard

[Insert screenshot of model performance charts including accuracy comparison, confusion matrix, and feature importance]

**Appendix B: Dataset Sample**

Appendix B-1: Sample rows from the enhanced_patient_data.csv dataset (1,830 records)

[Insert table showing 10 representative rows from the dataset including edge cases]

**Appendix C: Code Snippets**

Appendix C-1: Gradient Boosting model training with tuned hyperparameters

Appendix C-2: Healthcare recommendation algorithm (condition-to-specialist mapping)

Appendix C-3: Hospital contact database structure (sample entries for 5 districts)

**Appendix D: Turnitin Similarity Report**

The final report was submitted through Turnitin for plagiarism verification. The similarity index was confirmed to be within the acceptable threshold of 15% as required by SLIIT academic regulations.